

Micro-Lending & Peer Credit Risk Assessor

Functional and Non-Functional Requirements

Functional Requirements (FR)

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier (Low/Medium/High) is accurately assigned. Fail: Incomplete financial profile generates a credit risk warning.	Core functionality for assessing borrower risk.
FR-002	The system shall allow a borrower to submit the financial information required for credit risk assessment.	High	Pass: Borrower can submit all required financial details and the system confirms successful submission. Fail: Missing required information prevents submission.	Required to calculate the borrower's risk.
FR-003	The system shall allow a peer lender to view the borrower's credit risk rating before lending.	High	Pass: Lender can view the borrower's Low/Medium/High risk rating before approving a loan. Fail: Rating is unavailable or incomplete.	Helps lenders make informed lending decisions.
FR-004	The system shall generate a structured repayment schedule for an	High	Pass: A repayment schedule containing payment dates and amounts is generated after loan approval.	Supports organized repayment of the loan.

	approved micro-loan.		Fail: No schedule is generated.	
FR-005	The system shall trigger an automated warning when a borrower's credit risk is classified as High.	High	Pass: A warning is generated whenever the calculated risk tier is High. Fail: No warning is generated for a High-risk borrower.	Helps prevent risky lending decisions.

Non-Functional Requirements (NFR)

ID	Description	Priority	Acceptance Criteria	Rationale
NFR-001	The system shall execute repayment transactions and EMI distribution inside database ACID transactions to prevent double-spending.	High	Pass: Concurrent transaction tests show no duplicate or inconsistent EMI distribution. Fail: Any double-spending or inconsistent transaction occurs.	Protects financial transaction integrity.
NFR-002	The system shall protect borrower financial information from unauthorized access.	High	Pass: Unauthorized users cannot access borrower financial information. Fail: An unauthorized access attempt successfully retrieves financial data.	Financial information is sensitive and must be protected.